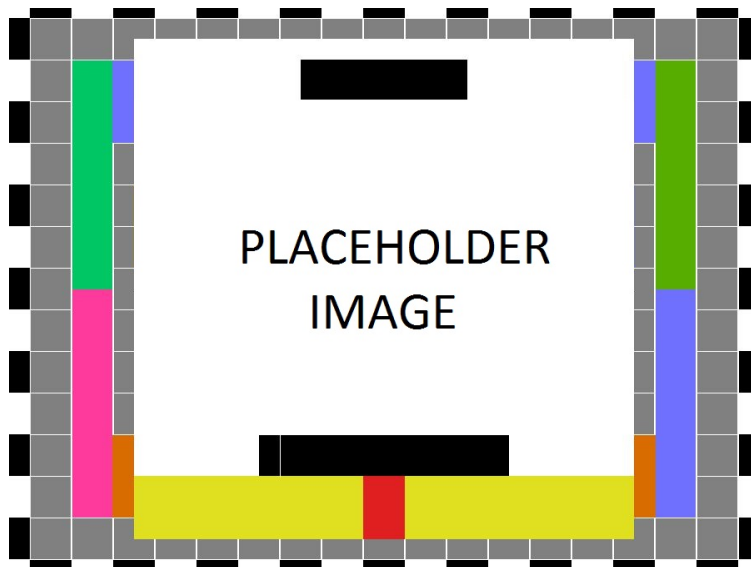


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**PROJECT CHARTER
CSE 4316: SENIOR DESIGN I
FALL 2026**



**TEAM TRAILBLAZER
TERRASCOUT AUTONOMOUS GROUND ROVER**

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REVISION HISTORY

Revision	Date	Author(s)	Description
0.1	09.02.2026	GH	document creation
0.2	09.06.2026	AT, GH	complete draft
1.0	09.15.2026	AT, GH, CB	official release

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1 PROBLEM STATEMENT

Utility-scale solar farms span hundreds of acres and thousands of photovoltaic (PV) panels. Undetected panel faults — cracked cells, soiling, and thermal hotspots — silently reduce energy yield and can become fire hazards. Today these sites are inspected manually on foot or with piloted drones, which is slow, labor-intensive, weather-limited, and inconsistent between inspectors. The result is that faults often go unaddressed for weeks. What is needed is a way to survey a PV field frequently, consistently, and without tying up skilled staff, so that faults are found and fixed while the lost energy is still small.

2 METHODOLOGY

Team Trailblazer will build TerraScout Autonomous Ground Rover, a small four-wheeled autonomous ground rover that patrols the drivable lanes between PV rows. Following a set of GPS waypoints, the rover captures visual and thermal imagery of the panels, flags likely faults on-board, and produces a GPS-tagged inspection report at the end of each run. When it detects a low battery or an obstacle it cannot pass, it returns to its charging dock autonomously. In short: we are building a robot that drives itself around a solar farm and tells the operator which panels need attention.

3 VALUE PROPOSITION

For the sponsor (a regional utility's O&M group), the rover converts a costly, irregular manual task into a repeatable automated one, catching yield-robbing faults earlier and reducing truck-rolls to healthy panels. For the university and instructors, the project exercises the full systems-engineering life cycle across embedded hardware, autonomy software, and computer vision, producing a demonstrable prototype and a reusable mobile-robot platform for future teams.

4 DEVELOPMENT MILESTONES

- Project Charter first draft - September 2026
- System Requirements Specification - October 2026
- Architectural Design Specification - November 2026
- Demonstration of tele-operated drive & sensor capture - December 2026
- Detailed Design Specification - February 2027
- Demonstration of waypoint navigation - March 2027
- Demonstration of on-board fault detection - March 2027
- CoE Innovation Day poster presentation - April 2027
- Demonstration of autonomous dock/return - April 2027
- Final Project Demonstration (full autonomous patrol) - May 2027

5 BACKGROUND

The sponsor operates several solar sites and currently contracts periodic drone thermography, typically once or twice a year. Between contracted flights the panels go uninspected. The sponsor approached the CSE Senior Design program seeking a lower-cost, higher-frequency ground-based alternative that their own field staff could operate without a licensed drone pilot. The team has no prior relationship with the sponsor; requirements were gathered through two scoping interviews.

6 RELATED WORK

Aerial PV inspection with thermographic drones is well established [1], but requires certified pilots and favorable weather, and does not run unattended. Commercial autonomous ground robots for solar exist but are priced for large operators and are closed platforms. Academic mobile-robot navigation using ROS, GPS-RTK, and visual fault detection is mature enough to adapt. TerraScout Autonomous Ground Rover targets the gap: an affordable, operator-runnable ground platform for frequent inspection. (Add at least five fully-documented IEEE-format references in the final version.)

7 SYSTEM OVERVIEW

At a high level the rover comprises a sensing front end (RGB + thermal cameras, GPS, IMU, ultrasonic bumpers), an on-board compute module running the autonomy and detection software, and a mobility base (brushless drive motors, battery, motor controllers). The operator interacts through a laptop ground-station web page over Wi-Fi to start runs and review reports; a charging dock provides autonomous recharge. See the Architectural Design Specification for the layered decomposition.

8 PROJECT MANAGEMENT APPROACH

The team follows an iterative Scrum model [1,2] consistent with ISO/IEC/IEEE 16326:2019 [3] within the life-cycle frameworks of ISO/IEC/IEEE 12207:2017 [4] and 15288:2023 [5]. Two-week sprints run across CSE 4316 (requirements/design) and CSE 4317 (build/test). Work is estimated in task-hours and tracked on a shared board; progress is measured by sprint burndown and requirement/test coverage. A burndown trending above the ideal line for three consecutive days triggers scope re-negotiation with the Product Owner. Requirement changes are approved by the Product Owner in consultation with the sponsor; all work products are version- controlled in Git with tagged document releases.

9 ROLES & RESPONSIBILITIES

Sponsor point of contact: utility O&M lead. Product Owner: Grace Hopper. Scrum Master: Alan Turing (rotating each semester). Areas of responsibility — Perception/CV: Ada Lovelace; Navigation/autonomy: John Von Neumann; Embedded platform/power: Charles Babbage; Ground station/reporting: Grace Hopper; Integration/test: Alan Turing.

10 COST PROPOSAL

Major expenses are the compute module, the thermal camera (the single largest line item), drive motors and controllers, battery, and chassis materials. Funding is the default CSE department stipend supplemented by a sponsor hardware contribution; the thermal camera may be loaned by the sponsor to stay within budget.

11 FACILITIES & EQUIPMENT

Development uses the Senior Design lab (soldering, bench power, 3D printing). Field testing uses a university parking area mocked up with panel stand-ins, and, later, a supervised visit to a sponsor solar site. RTK-GPS base station and a fenced outdoor test area are required by the third sprint.

12 ASSUMPTIONS

- A fenced outdoor testing area will be available by the 3rd sprint.
- The sponsor will loan a compatible thermal camera by the 4th sprint.
- Panel rows provide a drivable, reasonably level lane ($\leq 10\%$ grade).
- RTK-GPS correction will be available at the test site.
- The ground-station laptop and rover share a private Wi-Fi network.

13 CONSTRAINTS

- Final prototype demonstration must be completed by May 1st, 2027.
- Total development cost must not exceed \$800 excluding loaned equipment.
- The rover must weigh under 25 kg for safe two-person handling.
- Field testing only during daylight and dry weather.
- All captured imagery of sponsor sites must be sponsor-approved before public release.

14 RISKS

Risk description	Probability	Loss (days)	Exposure (days)
Thermal camera loan delayed	0.40	20	8.0
Outdoor RTK test area unavailable	0.30	14	4.2
Autonomy integration harder than estimated	0.50	10	5.0
Motor controller lead time (overseas)	0.20	18	3.6
Weather blocks field test windows	0.35	8	2.8

Table 2: Highest-exposure project risks

15 DOCUMENTATION & REPORTING

Documentation deliverables (Charter, SRS, ADS, DDS, Test Plan) are maintained in Git and released at the ends of their planning sprints. Recurring artifacts — product/sprint backlogs, sprint goals, burn-down charts, individual sprint reports, retrospectives, and engineering notebooks — are produced each sprint. Closeout materials include the rover prototype, a project poster, a public web page, a demo video, documented source code (MIT license), wiring schematics, chassis CAD (STEP), deployment scripts for the ground station, and an operator manual.

REFERENCES

- [1] K. S. Rubin, *Essential Scrum: A Practical Guide to the Most Popular Agile Process*, 1st ed. Addison-Wesley Professional, 2012.
- [2] K. Schwaber and J. Sutherland, “The Scrum Guide: The Definitive Guide to Scrum: The Rules of the Game,” 2020.

- [3] ISO/IEC/IEEE, “ISO/IEC/IEEE 16326:2019 — Systems and software engineering — Life cycle processes — Project management,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2019.
- [4] —, “ISO/IEC/IEEE 12207:2017 — Systems and software engineering — Software life cycle processes,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2017.
- [5] —, “ISO/IEC/IEEE 15288:2023 — Systems and software engineering — System life cycle processes,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2023.